Numerical verification of the refined Mazur–Tate conjecture in ranks 2 and 3, with an elementary formula for the Mazur–Tate pairing

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Abstract

Let E/Q be an elliptic curve of Mordell–Weil rank r and, for a prime m of good reduction, let θ_m ∈ Z[(Z/m)^x/±1] be the Mazur–Tate theta element built from plus modular symbols. The refined conjecture of Mazur and Tate (1987), in the formulation of Burns–Kurihara–Sano and Ota, predicts that θ_m lies in the r-th power of the augmentation ideal I and prescribes its image in I^r/I^{r+1} as a Tamagawa-type factor times the discriminant of the Mazur–Tate height pairing. We report a numerical verification of this leading-term prediction, with both sides computed independently, for nine elliptic curves of rank 2 (389a1, 433a1, 446d1, 571b1, 655a1, 709a1, 916c1, 1088j1, 1132a1) and one of rank 3 (5077a1), in about 700 non-trivial checks at prime levels m, with no failures; the curves were chosen to separate the roles of the Tamagawa numbers, the real components and the torsion, and an independent re-implementation with the constant predicted in advance confirmed the identity on seven further curves (237 non-trivial levels), and once the normalizations (period, sign of the modular symbol, orientation of the pairing, torsion) are made explicit the identity holds exactly in Ota's form, including his sign (−1)^{ν(m)}, assuming the standard normalization of the Artin map (uniformizer ↦ arithmetic Frobenius) used by Tan, which we have not checked in Ota's text; an independent re-implementation with the constant predicted in advance also confirms the identity in rank 1 (eight curves, 533 levels) and in rank 0 (eight curves, 249 levels), the only place where the factor #Ш is tested with #Ш > 1 (571a1, #Ш = 4; 681b1, #Ш = 9); in all, about 1700 non-trivial prime levels on 33 curves of ranks 0–3. At composite level $S = m_1m_2$ we verify Ota's inclusion–exclusion form of the conjecture for 389a1, including the mixed coefficient and the sign (−1)^{ν(S)}. The pairing side is computed by an elementary recipe: for R = c_Tam·#E(F_m)·P in the formal group at m, the diagonal value ⟨R,R⟩_m is the class of t(R)/d(R) in $F_m^x/\pm1$, where t = −x/y and d^2 is the denominator of x(R). We derive this recipe from the biextension definition of the pairing, assuming the Néron-model description of the canonical local splittings (as in Tan 1995; we have not had access to Mazur–Tate 1983), obtaining the closed formula ⟨R,S⟩ = [γ/A(R+S)] with γ the constant term of the primitive line through R and S and A the numerator of the x-coordinate; in particular $\langle R,R\rangle = [A(R)]^{-1}$. The elementary integrality step this requires is proved for every prime by an integral identity (Lemma in §2), so the only remaining hypothesis is that identification of the splittings. We know of no earlier numerical verification of the refined conjecture in rank 3, nor of one in rank 2 computing the pairing from coordinates (Portillo-Bobadilla 2019 does so in rank 1, by a denominator cross-ratio with an auxiliary point going back to Mazur–Tate (1987), with which our closed form agrees); Bley–Macías Castillo (2022) verified the equivalent 3-adic eTNC statement for 389a1, 433a1 and 446d1 at a handful of primes via Selmer groups and Hilbert symbols. As a secondary result we verify, in 20 cases over three curves and p ∈ {3,5}, that the Kolyvagin derived class κ_ℓ attached to a rank-2 curve and a Heegner field spans exactly the line $\ker(E(Q)/p \to E(F_\ell)/p)$, with an explicit nonvanishing criterion — a statement implicit in Howard and Mazur–Rubin and stated in an unpublished draft hosted on W. Stein's website; the nonvanishing part was tested in two cases by Jetchev–Lauter–Stein (2009), with which we agree, and the direction had not been tested. All computations are in PARI/GP 2.17 and are reproducible from the accompanying code.

1. Setup and notation

Throughout, E/Q is an elliptic curve of conductor N given by a minimal Weierstrass model, with Mordell–Weil group E(Q) of rank r and a fixed basis $P_1, \dots, P_r$ of E(Q)/E(Q)tors (saturated). We write $c_{Tam} = \prod_{\ell \mid N} c_\ell$ for the product of Tamagawa numbers and $c_\infty \in \{1, 2\}$ for the number of connected components of E(R).

Modular symbols. Let x^+ be the plus modular symbol of E, normalized as in PARI's `msfromell1`: for a rational cusp a/m, $[a/m]^+ := x^+(\{a/m\} - \{\infty\}) \in Q$ is the real part of $2\pi i\int$ from a/m to $i\infty$ of f(z)dz, divided by ω_1, the least positive real period of the Néron lattice (the real period over the connected component of E(R)). For the curves below, $[a/m]^+ \in Z$ when $c_\infty = 2$ and $[a/m]^+ \in \tfrac{1}{2}Z$ when $c_\infty = 1$.

Theta elements. For a prime m ∤ N let $G_m = (Z/m)^x$ and $G_m^+ = G_m/\pm1$, cyclic of order n := (m−1)/2. Since $[-a/m]^+ = [a/m]^+$, the Mazur–Tate element

$$\theta_m \;=\; \sum_{a \in (\mathbb{Z}/m)^\times} [a/m]^+ \, \sigma_a \;\in\; \mathbb{Q}[G_m^+]$$

is well defined. Fixing a primitive root g mod m and writing σ for the image of σ_g, we identify $Z[G_m^+] = Z[\sigma]/(\sigma^n - 1)$ and set x = σ − 1, so the augmentation ideal is I = (x). We expand $\theta_m = \Sigma_k a_k x^k$ with $a_k = \Sigma_t c_t C(t, k)$, where c_t is the coefficient of σ^t. The relation $(1+x)^n = 1$ gives $n x \equiv -C(n,2) x^2 - C(n,3) x^3 - \dots \pmod{I^n}$, so for $k \geq 1$ one has $I^k/I^{k+1} \cong Z/n$, and the class of θ_m in I^r/I^{r+1}, assuming $\theta_m \in I^r$, is obtained from $(a_0, \dots, a_r)$ by r successive reductions: if $a_j = n \cdot k_j$, replace a_{j-1+i} by $a_{j-1+i} - k_j\, C(n,i)$ for $i \geq 2$. We write $c_r(m) \in Z/n$ for the resulting class. The condition "$\theta_m \in I^r$" is exactly that $a_0 \equiv 0$ and that each intermediate reduced coefficient is $\equiv 0 \pmod n$; that $\theta_m \in I^r$ (the "order of vanishing" half of the conjecture) is a theorem of Ota (2018, Thm. 1.2) for the Z_p-parts with p ≥ 5 (2 and 3 inverted), E without complex multiplication and further hypotheses; our verification is over Z, including the 2- and 3-parts; we verify it in every case below.

The refined conjecture at prime level. In the form of Burns–Kurihara–Sano (2021, eqs. (8)–(9)) and Ota (2018, Conj. 2.4) for S = {m}:

$$\theta_m \;\equiv\; (-1)^{\nu(m)}\, \#Ш(E/\mathbb{Q})\; J_m\; R_F \pmod{I^{r+1}}, \qquad J_m = \#\operatorname{coker}\Big(E(\mathbb{Q}) \to E(\mathbb{F}_m) \oplus \bigoplus_{\ell \mid N} E(\mathbb{Q}_\ell)/E_0(\mathbb{Q}_\ell)\Big)$$

where R_F = det(⟨x_i, y_j⟩) is the discriminant of the Mazur–Tate pairing ⟨·,·⟩: E(Q) × E_S(Q) → G_m⁺ ≅ I/I², with x_i a basis of E(Q) and y_j a basis of the "S-Selmer" sublattice E_S(Q) = kernel of the map defining J_m. For the curves considered, #Ш = 1 (analytically) and E(Q)_tors = 0, except 1088j1 with E(Q)_tors ≅ Z/2.

2. An elementary formula for the Mazur–Tate pairing at a prime level

The Mazur–Tate pairing is defined idèlically via a biextension (Mazur–Tate 1983, 1987; see also Tan 1995, Def. 2.6–2.8). Its definition does not yield a closed formula directly; earlier numerical work computed it as a cross-ratio of denominators with an auxiliary point (a formula of Mazur–Tate 1987, p. 747, used by Portillo-Bobadilla 2019, eq. (18), in rank 1 — see the comparison at the end of this section), in Darmon's Heegner-point setting (Séguin 2014), or via Selmer groups and Hilbert symbols (Bley–Macías Castillo 2022). We use the following recipe, suggested by the analogy with the Mazur–Stein–Tate formula h_p(P) = (1/p)·log_p(σ_p(P)/d(P)) for cyclotomic p-adic heights:

Recipe. Let R ∈ E(Q) lie in the formal group $E_1(\mathbb{Q}_m)$, write x(R) = a/d² in lowest terms with d > 0, and let t(R) = −x(R)/y(R) be the formal parameter. Then v_m(t) = v_m(d) ≥ 1, the ratio t(R)/d(R) is an m-adic unit, and we set

$$h(R) \;:=\; \log_g\Big[\,t(R)/d(R) \bmod m\,\Big] \;\in\; \mathbb{Z}/n, \qquad \langle R,R\rangle_m \;:=\; 2\,h(R) \;=\; \log_g\big[(t/d)^2\big] \;=\; -\log_g\big[A(R)\big],$$

where x(R) = A(R)/d² and the last equality uses B² ≡ A³ (mod m) for y(R) = B/d³, since t/d = −A/B exactly. We call h the *half-pairing*: it is a quadratic form on E_S(Q) and the pairing is its polarization, ⟨R₁,R₂⟩ := h(R₁+R₂) − h(R₁) − h(R₂), which gives back 2h(R) on the diagonal; no division by 2 is ever needed. The pairing matrix of a basis R₁, …, R_r of a sublattice is H := (⟨R_i,R_j⟩), with H_ii = 2h(R_i). The m-adic sigma function does not appear because σ_m(t) = t·(1 + O(t)) and t ≡ 0 (mod m), so σ_m(R)/d(R) ≡ t(R)/d(R) (mod m). The justification of these formulas is in the paragraph "The local computation" below.

Points in the formal group. For an arbitrary basis P_i of E(Q) we take R_i = M·P_i with M = c_Tam·#E(F_m). Multiplication by #E(F_m) kills the reduction mod m, and multiplication by c_Tam puts the point into $E_0(\mathbb{Q}_\ell)$ at every bad ℓ, so R_i lies in the lattice E_S(Q) of §1 (this is the only place where the Tamagawa numbers enter, and §4 shows that omitting c_Tam breaks the identity for 446d1, the one curve on which the omission was tested). The coordinates of R_i are large rationals (roughly M²·ĥ(P_i)/log 10 digits) but PARI handles them exactly for m up to a few thousand.

Status of the recipe: a consequence of Tan's definition, up to one local computation. We have not had access to Mazur–Tate (1983, 1987), but Tan (1995, §2) gives the definition in a form that can be compared with the recipe. Tan's pairing E(K) × E_S(K) → G is the composite of a pairing with values in C_S = ∏ C_v (Def. 2.6) with the Artin map C_S → G (Def. 2.8), where the local groups are: at a tamely ramified place v ∈ S, A_v = E(K_v), B_v = E_1'(K_v) (points with trivial reduction) and C_v = $K_v^{\times}/(1 + \pi_v\, O_v) \cong \mathbb{Z} \times k_v^{\times}$; at an unramified non-archimedean v, B_v = E_0(K_v) and C_v = $K_v^{\times}/O_v^{\times} \cong \mathbb{Z}$, the pairing being the Néron local symbol; at a real place unramified in L/K, C_v = 1. For K = Q, S = {m}, L = $\mathbb{Q}(\mu_m)^+$ this says three things. (i) The domain of the second argument is exactly the lattice E_S(Q) = {R : R ∈ $E_1(\mathbb{Q}_m)$, R ∈ $E_0(\mathbb{Q}_\ell)$ for ℓ | N} that the recipe requires — the hypotheses of the recipe are the hypotheses of the definition. (ii) The Artin map kills the uniformizer m at the place m and the subgroup 1 + mZ_m, so only the unit part of the local value survives, as a class in $\mathbb{F}_m^{\times}/\pm 1 = G_m^+$; and at ℓ ≠ m it sends ℓ^k to $\mathrm{Frob}_\ell^k = [\ell^k]$. Hence ⟨R,R⟩ = Art($\psi_m$(p)) · $\prod_l$ [ℓ]^ψ_l(p) for any point p of the biextension over (R,R), the product being independent of p because the global units are killed (φ($Q^{\times}$) = 0). (iii) The splitting ψ_m is the unique one on the formal group (Hom(E_1, Hom(E_1, C_m)) = 0 since E_1 is pro-m and C_m has no m-torsion), so it coincides with the reduction mod (1 + mZ_m) of the Mazur–Tate m-adic σ-function, which is t·(1 + O(t)); and the Néron symbols at ℓ | d sum to ord_l of the denominator. The recipe t/d is precisely this product, read as the quadratic form whose polarization is the pairing; t/d is the half-pairing h, whose polarization is the pairing; the local computation below shows that the pairing itself is [(t/d)²] on the diagonal. What remains to make this a proof is the local computation at m and ℓ for one explicit representative p over (R,R), with the $Q^{\times}$-ambiguity removed; we have not carried it out. The orientation of the Artin map on local units (u ↦ σ(u) or σ(u^{-1})) decides whether Tan's pairing is ours or its inverse; we fix the orientation that makes the formulas below hold, and §4(b) shows from the data that Tan's is the inverse.

The local computation. We then carried out the computation that the previous paragraph left open; it is short. Realize the Poincaré biextension of E × E by G_m in the usual way: a point of P over (a, b) is a non-zero rational function f on E with divisor (a+b) + (O) − (a) − (b), and the G_m-action is scaling. For a = R, b = S in E_S(Q) take

f = (x − x(R+S)) / ℓ(x, y), ℓ = the line through R and S (the tangent line if R = S),

which has the right divisor, and scale f by a rational number so that both the numerator d(R+S)²x − A(R+S) and the denominator ℓ_0 = αy + βx + γ (α, β, γ ∈ Z, gcd = 1) are primitive integral forms. Here x(R+S) = A(R+S)/d(R+S)² in lowest terms. The canonical local splittings of Mazur–Tate, in the description through the Néron model, are: at an unramified place ℓ (B_ℓ = $E_0(\mathbb{Q}_\ell)$, C_ℓ = Z) the order of f along the identity component of the special fibre — which is 0 for a primitive f — and at the tamely ramified place m (B_m = $E_1(\mathbb{Q}_m)$, C_m = $\mathbb{Q}_m^{\times}/(1+m\mathbb{Z}_m)$) the m-adic order together with the reduction of f to the special fibre: since R, S, R+S all reduce to O, the reduction $\bar{f}$ has trivial divisor and is a non-zero constant in $\mathbb{F}_m^{\times}$, and this constant is ψ_m(f). The reduction is immediate from the primitive forms: d(R+S) ≡ 0, so the numerator reduces to −A(R+S); and since R, S ∈ $E_1(\mathbb{Q}_m)$ the coefficients α, β of the primitive line are divisible by m (the line through two points of E_1 has slope and intercept with m-adic poles of order ≥ 1 and ≥ 3, so after clearing denominators the constant term is the unit), and the denominator reduces to the constant γ. Hence ψ_m(f) = −A(R+S)/γ ∈ $\mathbb{F}_m^{\times}$, all other components are trivial, and by Artin reciprocity

⟨R, S⟩ = [γ / A(R+S)] ∈ $\mathbb{F}_m^{\times}/\pm 1 = G_m^+$,

where we fix the orientation of the Artin map on local units so that this holds (the other orientation inverts every value). This is a closed formula for the Mazur–Tate pairing, valid for all R, S ∈ E_S(Q) with R, S, R+S ≠ O. It remains to compare it with the recipe. For R = S, writing x(R) = A/d², y(R) =

B/d^3, the tangent-line constant is $\gamma = (BW - \Lambda A)/c_L$ with $W = 2B + a_1Ad + a_3d^3$, $\Lambda = 3A^2 + 2a_2Ad^2 + a_4d^4 - a_1Bd$ and c_L the content of the line; modulo m ($d \equiv 0$, and $B^2 \equiv A^3$ from the Weierstrass equation) this is $\gamma \equiv -A^3/c_L$, while $A(2R) \equiv A^4/\delta^2$, where δ is the cancellation in the duplication formula $x(2R) = N_4/(d^2W^2)$, $N_4 := A^4 - b_4A^2d^4 - 2b_6Ad^6 - b_8d^8$, defined by $\delta := |dW|/d(2R)$ (so $\delta^2 = N_4/A(2R)$ when $A(2R) \neq 0$; see the Lemma below). So $\langle R,R\rangle = [-\delta^2/(c_L \cdot A)]$, and the recipe gives $\langle R,R\rangle = 2h(R) = [(t/d)^2] = [A/B]^2 = [A]^{-1}$ (because $t/d = -A/B$ exactly and $B^2 \equiv A^3$). The two agree if and only if $\delta^2 \equiv \pm c_L \pmod m$. The content c_L is always 1: since $BW - \Lambda A \equiv -A^3 \pmod d$ is coprime to d, $c_L = \gcd(W, \Lambda)$, and a prime ℓ divides both $W = d^3 \cdot \partial F/\partial y(R)$ and $\Lambda = -d^4 \cdot \partial F/\partial x(R)$ (F the Weierstrass equation) only if $\bar{R}$ is a singular point of the reduction, which $R \in E_0(Q_l)$ excludes. For δ we have the following, which holds at every prime, $\ell = 2$ included, and uses no formal group.

Lemma (the cancellation δ). Let E/Q be given by a Weierstrass equation $F = 0$ with integral coefficients and let $R \in E(Q)$ with $2R \neq O$, $x(R) = A/d^2$, $y(R) = B/d^3$ and $x(2R) = A(2R)/d(2R)^2$ in lowest terms. With W, Λ, N_4 as above and $\delta := |dW|/d(2R)$: δ is an integer, $\delta^2 = \gcd(N_4, d^2W^2)$, and for every prime ℓ, $\ell \mid \delta$ if and only if $R \notin E_0(Q_l)$. In particular $\delta = 1$ for every $R \in E_S(Q)$ of infinite order.

Proof. Substituting the tangent slope $\lambda = -F_x/F_y = \Lambda/(dW)$ into $x(2R) = \lambda^2 + a_1\lambda - a_2 - 2x$ and clearing denominators gives the identity in $Z[a_1, \ldots, a_6, A, B, d]$

$$N_4 = \Lambda^2 + a_1 d\Lambda W - (a_2d^2 + 2A)W^2 + (b_2d^2 + 8A)\cdot G,\ G := d^6 \cdot F(A/d^2, B/d^3),$$

and $G = 0$ at R, so $x(2R) = N_4/(d^2W^2)$. Since $A(2R)$ and $d(2R)$ are coprime, $A(2R) \cdot d^2W^2 = N_4 \cdot d(2R)^2$ forces $d(2R) \mid dW$; with $\delta = |dW|/d(2R)$ this gives $N_4 = \delta^2 A(2R)$ and $d^2W^2 = \delta^2 d(2R)^2$, hence $\delta^2 = \gcd(N_4, d^2W^2)$. Now let $\ell \mid \delta$, so $\ell \mid N_4$ and $\ell \mid dW$. If $\ell \mid d$ then $N_4 \equiv A^4 \pmod \ell$ with $\ell \nmid A$, impossible; so $\ell \nmid d$ and $\ell \mid W$, and the identity gives $N_4 \equiv \Lambda^2 \pmod W$, so $\ell \mid \Lambda$. As d is an ℓ-adic unit, $W = d^3F_y(R)$ and $\Lambda = -d^4F_x(R)$ both vanish mod ℓ, i.e. $\bar{R}$ is singular: $R \notin E_0(Q_l)$. Conversely, if $R \notin E_0(Q_l)$ then $\bar{R} \neq O$, so $\ell \nmid d$, and $\ell \mid W$, $\ell \mid \Lambda$; the identity gives $\ell \mid N_4$, so $\ell \mid \gcd(N_4, d^2W^2) = \delta^2$. For $R \in E_S(Q)$, $R \in E_0(Q_l)$ at every ℓ (bad ℓ by definition, good ℓ trivially), so $\delta = 1$. ■

The identity was also checked symbolically (PARI; the two differences reduce to 0), and the lemma numerically on 1050 points of 52 curves chosen to cover every Kodaira type at 2, with no failures; it does not need the model to be minimal, and for points outside $E_0(Q_2)$ it explains the values $\delta = 2$ and 8 found on 446d1 and 1088j1. (Earlier drafts had a formal-group argument valid only for odd ℓ; the lemma was found in an isolated session and verified by the registration session.) Numerically (chkg2.gp), $\delta = c_L = 1$ and $\gamma/A(2R) \equiv \pm(t/d)^2$ at all 349 points $R_i = M \cdot P_i$ with $7 \le m \le 113$ on six curves (389a1, 446d1, 1088j1, 916c1, 655a1, 5077a1), and in a separate sweep (gsweep.py) $\delta = c_L = 1$ at all 3631 points $c_{Tam} \cdot k \cdot P$, $1 \le k \le 8$, P running over the generators, on the 453 curves of positive rank and conductor ≤ 400. For $R \neq S$ the same computation with the chord gives $\langle R,S\rangle = [\gamma_{R,S}/A(R+S)]$, and at 129 pairs (R_1, R_2) over five curves this equals $[(t/d)(R+S)/((t/d)(R)(t/d)(S))]$, i.e. exactly the polarization $h(R+S) - h(R) - h(S)$ of the recipe — with no 2-torsion ambiguity, which the polarization of a quadratic form alone would leave when n is even. Scripts: `biext.gp`, `chkg2.gp`, `chord.gp`, `gsweep.py`.

Comparison with Portillo-Bobadilla's formula. After the above was written we found that Portillo-Bobadilla (2019, §3.2, eq. (18) and Prop. 3.1) computes the same pairing at a prime $p \nmid N$, for his rank-1 verification, by a cross-ratio of denominators with an auxiliary point P': $g(P, Q, P', p) = d(P'+P) \cdot d(P'+Q) / (d(P') \cdot d(P'+P+Q))$ (mod p), with $d(T)$ the square root of the denominator of $x(T)$ and Q in the formal group, which he shows is independent of P' and bimultiplicative. Portillo-Bobadilla states that g is the function defined by Mazur and Tate (1987, p. 747; his reference [12] is the Duke paper), which we have not been able to consult; the auxiliary-point formula is therefore Mazur–Tate's own, and his contribution is its use in the numerical test. This is the Néron-symbol expression of the biextension value with the representative $(P'+P) - (P')$ in place of our $(P+Q) + (O) - (P) - (Q)$. We checked numerically that his $g(R, S, P', m)$ equals our $[\gamma/A(R+S)]$ up to sign (i.e. exactly in $F_m^x/\pm 1$) on 389a1 at $m = 23, 29, 31, 37, 41$, for $R = S$ and $R \neq S$ and for three choices of P' (15/15; script `portillo.gp`). Our formula is therefore not new as a way of evaluating the pairing from coordinates; what is new here is the auxiliary-point-free closed form — in particular $\langle R,R\rangle = [A(R)]^{-1}$, the identification with the half-pairing t/d, and the Lemma — and its use in ranks 2 and 3. The agreement is also an independent check of the local computation above against a published formula.

In summary: granting the Néron-model description of the canonical local splittings (which is how Mazur–Tate 1983 construct them and how Tan 1995 uses them), the Mazur–Tate pairing on $E_S(Q) \times E_S(Q)$ at prime level is given exactly by the recipe of this section, with $\delta = c_L = 1$ on $E_S(Q)$ proved above; the one remaining hypothesis is that description of the splittings. The diagonal takes the memorable form $\langle R,R\rangle = [A(R)]^{\mp 1}$: the Mazur–Tate self-pairing of a point of $E_S(Q)$ is the class of the numerator of its x-coordinate.

3. The identity tested

Let $I_m := [E(Q) : E_S(Q)]$ be the size of the image of $E(Q)$ in $E(F_m) \oplus \bigoplus_{\ell \mid N} E(Q_\ell)/E_0(Q_\ell)$, so that $J_m = M/I_m$ with $M = c_{Tam} \cdot \#E(F_m)$ as above. Since the pairing is bilinear with $E_S(Q)$ in one argument, $\langle x_i, y_j\rangle = M^{-1}\langle M x_i, y_j\rangle$, and since $M \cdot E(Q) \subset E_S(Q)$ has index M^r/I_m, a change of basis gives

$$J_m\, R_F \;=\; \frac{M}{I_m} \cdot \frac{I_m}{M^{2r}} \cdot \det\big(\langle R_i, R_j\rangle_m\big) \;=\; \frac{\det\big(\langle R_i, R_j\rangle_m\big)}{M^{2r-1}}, \qquad R_i = MP_i.$$

The index I_m therefore cancels, and the prediction of §1 becomes, for $\#Ш = 1$ and up to a constant u_E depending only on normalizations,

$$\boxed{M^{2r-1}\, c_r(m) \;\equiv\; u_E \cdot \det_{r\times r}\big(\langle R_i, R_j\rangle_m\big) \pmod n, \qquad n = (m-1)/2}$$

In practice we test $2^r\, M^{2r-1}\, c_r(m) \equiv u_E \cdot \det H \pmod n$ with $H = (\langle R_i,R_j\rangle)$ the pairing matrix of §2; the factor 2^r on the left is a historical artefact of an earlier normalization and is simply absorbed into u_E (so $u_E = 2^r \times$ the constant of the boxed identity). Both sides depend on the

primitive root g: replacing g by g^k multiplies c_r by k^{−r} and det H by k^{−r}, so the identity is independent of g. The constant u_E must be the same for all m; we determine it from the first few primes and then test it on all others. The test is non-trivial at a prime m when (lhs, rhs) ≠ (0, 0); when gcd(M, n) > 1 the identity constrains c_r(m) only modulo n/gcd, and we flag these as weaker checks.

A first version of this computation used the factor N_m/I_m^2 ($N_m := \#E(F_m)$) instead of M^{-3} for r = 2; the two agree modulo 3 whenever $3 \nmid N_m$ (because $N_m^4 \equiv 1 \bmod 3$) but not modulo n, and the mod-3 test passed without revealing the error. We mention this because it illustrates why the full-level test is the one that carries information.

4. Results

All ten curves satisfy $\theta_m \in I^r$ at every prime level tested (the successive reductions of §1 never failed), and the boxed identity of §3 holds at every level with a constant u_E depending only on the curve.

Curve (Cremona)	r	N	c_Tam	c_∞	Basis P_i	u_E	prime levels m	non-trivial checks	of which gcd(M,n) > 1	failures
389a1	2	389	1	2	(−1,1), (0,0)	8	5 ≤ m ≤ 1500	221	106	**0**
709a1	2	709	1	2	saturated `ellrank` basis	8	m ≤ 700	111	58	**0**
446d1	2	446 = 2·223	2 ($c_2 = 2$)	2	(2,0), (1,0)	8	23 ≤ m ≤ 400	61	—	**0**
571b1	2	571	1	1	(1,−1), (0,−2)	4	23 ≤ m ≤ 330	54	—	**0**
5077a1	3	5077	1	2	(1,−1), (2,−1), (0,−3)	−16	23 ≤ m ≤ 330	49	—	**0**
655a1	2	655 = 5·131	2	1	(1,2), (3,2)	4	23 ≤ m ≤ 330	51	—	**0**
1132a1	2	1132 = 2^2·283	3	1	(1,1), (−1,−3)	4	23 ≤ m ≤ 330	45	—	**0**
916c1	2	916 = 2^2·229	3	2	(0,1), (2,1)	8	23 ≤ m ≤ 330	46	—	**0**
1088j1 (torsion Z/2)	2	1088 = 2^6·17	4	2	(−1,8), (7,16)	2	23 ≤ m ≤ 300	44	—	**0**
433a1	2	433	1	1	(0,1), (−1,0)	4	7 ≤ m ≤ 113	21	—	**0**

"Non-trivial" excludes levels where both sides vanish (15 of the 236 levels for 389a1, for instance). Levels dividing N are excluded. The curves 655a1, 1132a1, 916c1 and 1088j1 were chosen, after a first draft of this note, to separate the three invariants that could enter the constant: $c_\infty = 1$ with c_Tam = 2 and 3 (655a1, 1132a1), $c_\infty = 2$ with odd c_Tam (916c1), and non-trivial torsion (1088j1). The result is that u_E depends on c_∞ and on the torsion, and not at all on the Tamagawa numbers (which are fully absorbed by the factor c_Tam in M):

$u_E = (-2)^r \cdot c_\infty / \#E(Q)_tors^2$ in all ten cases (8, 8, 8, 4, −16, 4, 4, 8, 2, 4).

Every factor, including the sign, is accounted for. (a) 2^r is the factor we put on the left-hand side of the test (§3) and cancels; in terms of the pairing matrix H the identity is $M^{2r-1}\, c_r(m) \equiv (-1)^r \cdot (c_\infty / \#E(Q)tors^2) \cdot \det H$. (b) The sign $(-1)^r$ — + for the rank-2 curves, − for the rank-3 curve, and − for rank 1 (eight curves, 533 levels, in the independent re-implementation of §4 below) — is the product of two conventions. First, the pairing: Tan's Artin map (his §1, Weil group $W_D = K^x\backslash A^x/U_D$) sends a uniformizer π_v to the Frobenius [v]; by global reciprocity a local unit u at m then corresponds to σ_u^{-1} (choose $a \in Z$ with $a \equiv u \bmod m$; the principal idèle a gives $\prod_{\ell} \mathrm{Frob}_\ell^{v_\ell(a)} = \sigma_a$, so $[u \text{ at } m] = \sigma_a^{-1}$), whereas our recipe uses $u \mapsto \sigma_u$. So Tan's pairing is the inverse of ours, $P := -H$, i.e. $\langle R,R\rangle = [A(R)]$ rather than $[A(R)]^{-1}$, and $\det P = (-1)^r \det H$, which removes the $(-1)^r$. Second, the theta element: PARI's plus symbol satisfies $x^+(\{0\} - \{\infty\}) = -L(E,1)/\omega_1$ (checked on seven rank-0 curves of both discriminant signs), whereas Mazur–Tate normalize $[0]^+ = +L(E,1)/\Omega^+$; so θ_m of §1 is $-c_\infty$ times the Mazur–Tate element θ_m^+, and this global −1 is exactly Ota's $(-1)^{v(m)} = -1$ at prime level — provided Ota's Artin map is normalized as Tan's (uniformizer ↦ arithmetic Frobenius), which is the standard choice but which we have not checked in his text; this is question 3 of our letter to him. An earlier draft of this paragraph said the discrepancy could not be removed by an orientation change; that was wrong because it took the sign of θ_m for granted, and the rank-1 data (which force a factor $(-1)^{r+1} = -(-1)^r$, not $(-1)^{r(r-1)/2}$) exposed it. The composite-level test below confirms the relative sign $(-1)^{v(S)-v(m)}$ as well. (c) The symbols x^+ of PARI are normalized by the least real period ω_1 of the Néron lattice, which is the full real period when $c_\infty = 1$ but only half of it when $c_\infty = 2$; and θ_m as defined in §1 is $2\cdot\Sigma\,[a/m]^+\,\sigma_a$ over $a \in G_m^+$, because both a and −a are summed. If instead the symbols are normalized by Ω^+_int := the generator of the real projection of the lattice — which is ω_1 when $c_\infty = 2$ and $\omega_1/2$ when $c_\infty = 1$, and is the normalization that makes $[a/m]^+$ an integer in all cases — and $\theta_m^+ := -\Sigma\,[a/m]^+\,\sigma_a$ over $a \in G_m^+$ (the sign from (b)), then $\theta_m = -c_\infty \cdot \theta_m^+$ and the factor c_∞ disappears. (d) The factor $1/\#E(Q)tors^2$ is the torsion normalization of the discriminant in Tan's Def. 2.13 ($\mathrm{disc}_S = |A_tors|^{-1}|B_tors|^{-1}$ det) combined with the torsion contribution to the index $[E(Q) : E_S(Q)]$ in J_m; it is the same $tors^2$ as in the Birch–Swinnerton-Dyer formula and in Burns–Kurihara–Sano's eq. (9). With these normalizations the identity reads, for all curves and ranks 0–3 (rank 0: eight curves and 249 levels in the re-implementation, including #Ш = 4 and 9; ranks 1–3: this note and the re-implementation),

$$\theta_m^+ \equiv (-1)^{v(m)} \cdot \#Ш \cdot J_m \cdot \mathrm{disc}_m(P) = -\#Ш \cdot \det P \,/\, (M^{2r-1} \cdot \#E(Q)tors^2) \pmod{I^{r+1}},$$

with $P = (\langle R_i,R_j\rangle)$ in Tan's orientation, $\langle R,R\rangle = [A(R)]$ (and #Ш = 1 for every curve of positive rank here),

which is Ota's Conj. 2.4 at prime level with no undetermined constant. The value u_E = 4 for 571b1 supersedes the value 2 quoted in an earlier draft, which we could not reproduce.

The Tamagawa test (446d1). With $R_i = \#E(F_m)\cdot P_i$, i.e. omitting c_Tam = 2, the identity fails at 28 levels of a run over 5 ≤ m ≤ 400. With $R_i = 2\cdot\#E(F_m)\cdot P_i$, so that $R_i \in E_0(Q_2)$ as the definition of E_S(Q) requires, it holds at 61 of 61 non-trivial levels with the same u = 8 as for the other $c_\infty = 2$ curves. This is the definition of J_m through the joint cokernel in Burns–Kurihara–Sano being confirmed in detail.

The vanishing levels. For 389a1 at $m \equiv 1 \pmod 9$ the mod-3 reduction of the identity was examined separately: the levels at which $\mathrm{ord}_I\,\theta_m$ jumps from 2 to 4 (m = 37, 163, 613, 757, 829, 937 among m ≤ 1200) are exactly the levels at which $\det\langle R_i,R_j\rangle \equiv 0 \pmod 3$. The frequency of such jumps (6 of the 20 levels $m \equiv 1 \pmod 9$ in the mod-3 run, m ≤ 1200) matches the probability that a random symmetric 2×2 matrix over F_3 is singular (9/27).

Jumps are always by an even amount (ord ∈ {2, 4, 6}), consistent with the alternating character of the second derived pairing (Bertolini–Darmon 1994, Thm. 3.18). At levels with 3 | #E(F_m), where the mod-3 identity is not directly applicable, the implication "leading term ≢ 0 mod p ⇒ p ∤ J_m" of Ota's theorem — which is proved only for p ≥ 5 and under further hypotheses, so that at p = 3 it is a consistency check and not a theorem — would force $c_2 \equiv 0 \pmod 3$ at m = 127, 487, 919, which is what occurs. The re-implementation checked the same implication at p = 5 and 7 on eight curves (m ≤ 3000; m ≤ 10000 for 389a1 and 709a1): 11 levels with p | J_m (8 at p = 5, 3 at p = 7), no violations. Nine of them satisfy all the hypotheses of Ota's theorem that we could check (no CM; p ∤ 6N·#E(F_p)·∏c_ℓ; E(F_m)[p] cyclic; $\bar{\rho}$_p surjective, by Serre's criterion for semistable curves and the absence of a rational p-isogeny); the other two (389a1 at m = 1831, where E(F_m)[5] is not cyclic, and 997b1 at p = 5, where 5 | #E(F_5)) are outside its range. No level with p ≥ 11 dividing J_m occurs in this range; such levels are structurally rare (p | (m−1)/2, p | #E(F_m) and non-surjectivity of E(Q) → E(F_m)/p at once).

Rank 3. For 5077a1 the class $c_3(m) \in I^3/I^4$ and the 3×3 determinant agree with u = −16 at all 49 non-trivial levels tested. The order-of-vanishing statement θ_m ∈ I^3 alone (tested mod 3 on 8 further levels m ≡ 1 mod 9) showed the leading coefficient vanishing mod 3 more often than in rank 2 (6 of 8), which the full-level computation explains as degeneracy of the 3×3 pairing mod 3 rather than any defect in the conjecture.

Composite levels (Ota's inclusion–exclusion). For S = m_1m_2 the conjecture (Ota, Conj. 2.4, which refers to Mazur–Tate 1987, (2.5)–(2.6), (3.1) for d_T and μ_{S,T}; we reconstructed μ_{S,T} from the norm relation rather than from that paper) reads, for $\tilde{\theta}_S$ the image of θ_S in Z[G_S^+] (Ota's notation), $\tilde{\theta}_S \equiv$ #Ш · Σ_{T | S} (−1)^{ν(T)} J_T η_r(μ_{S,T}(d_T)) in I_S^r/I_S^{r+1}, where d_T is the discriminant of the level-T pairing and μ_{S,T} lifts $\mathrm{Sym}^r(G_T^+)$ to $\mathrm{Sym}^r(G_S^+)$ and multiplies by φ(S/T). We push everything to the quotient $\bar{G} = Z/n_1 \times Z/n_2$ of G_S^+ ($n_i = (m_i - 1)/2$), where $I^2/I^3 \cong Z/n_1 \cdot x_1^2 \oplus Z/g \cdot x_1x_2 \oplus Z/n_2 \cdot x_2^2$ with g = gcd(n_1, n_2) (Passi's theorem $I^2/I^3 \cong \mathrm{Sym}^2(\bar{G})$), and the level-S pairing of R ∈ E_S(Q) is the pair of level-m_i recipes (h^1(R), h^2(R)). Writing H^k for the polarized matrix at m_k and $R_i = c_{Tam} \cdot \#E(F_{m_1}) \cdot \#E(F_{m_2}) \cdot P_i$, the discriminant is $\det(H^1x_1 + H^2x_2) = D_1x_1^2 + D_{12}x_1x_2 + D_2x_2^2$ with $D_k = \det H^k$ and $D_{12} = H^1{}_{11}H^2{}_{22} + H^2{}_{11}H^1{}_{22} - 2H^1{}_{12}H^2{}_{12}$. Two consequences, in rank 2: the terms $T = m_1, m_2$ contribute only to x_i^2, where the sum over T collapses (by the norm relation $\pi_{S/m_2}\theta_S = (a_{m_2} - \sigma - \sigma^{-1})\theta_{m_1}$, which is a theorem) to the statement that the x_1^2-coefficient of θ_S equals $(a_{m_2} - 2) \cdot c_2(m_1) \bmod n_1$; and the mixed coefficient receives only the term T = S, giving the one genuinely new prediction (a_{11} denotes the coefficient of x_1x_2 in the expansion of θ_S, after the reductions of §1 in each variable)

$$4\,M^3 \cdot a_{11}(\theta_S) \equiv -u_E \cdot D_{12} \pmod g, \quad M = c_{Tam} \cdot \#E(F_{m_1}) \cdot \#E(F_{m_2}),$$

with the same u_E = 8 as at prime level and the sign reversed, because ν(S) = 2. For 389a1 we tested all 28 pairs of primes $7 \le m_1 < m_2 \le 61$ with g ≥ 3 (code `mt_ota.gp`, `mt_ota_run.py`): $\theta_S \in I_S^2$ in every case; the x_1^2 and x_2^2 norm-relation checks hold in every case; and the mixed identity holds in every case, non-trivially (both sides ≠ 0) at 10 pairs with g ∈ {3, 5, 6, 7}, including (11, 31) mod 5 and (29, 43) mod 7, where the alternatives u = −8 (no sign change) and u = ±4, ±16 are excluded. The remaining pairs have $D_{12} \equiv 0 \equiv a_{11} \pmod g$ and are consistent but uninformative. The cost is in the points, not the symbols: R_i has about $(m_1m_2)^2 \cdot \hat{h}(P_i)$ nats, i.e. 10^5–10^6 digits for $m_i \le 61$, so m_i of a few hundred are out of reach of exact arithmetic without a p-adic shortcut. An earlier draft of this paragraph, written before the formula was implemented, said that the three second-order coefficients "do not follow a single-constant extrapolation of the prime-level identity"; that was correct and is explained by the T = m_i terms above.

Comparison with Bley–Macías Castillo at their levels. Bley and Macías Castillo (2022, §4.4) verify the 3-part of the equivariant Tamagawa number conjecture for E over the cubic subfield F of Q(ζ_ℓ) for 389a1 (ℓ = 7, 13, 43), 433a1 (ℓ = 7, 13, 31, 37) and 446d1 (ℓ = 19). Their statement (Thm. 2.1, condition (11)) is that L*/Reg lies in $Z_3[G]^x$, where Reg is built from the Néron–Tate regulator and a 2×2 matrix Ψ of Mazur–Tate pairing values ⟨P_i, P_j⟩ ∈ I(G)/$I(G)^2$ ≅ Z/3 (minus the Bertolini–Darmon pairing, computed from generalized 3-Selmer groups and Hilbert symbols in Magma); their published output (file ETNC_V.out on W. Bley's page) records only the verdict "ETNC is valid" per pair, not the values of Ψ or of L*, and their code requires det Ψ ≠ 0 (mod 3) as part of the verdict. The comparison we can make without Magma is therefore at the level of that determinant: in our data, det⟨R_i, R_j⟩_m (mod 3) is non-zero at all eight of their levels (389a1: 2, 1, 2 at m = 7, 13, 43; 433a1: 2, 2, 1, 1 at m = 7, 13, 31, 37; 446d1: 1 at m = 19), in agreement with the nonvanishing their verification certifies, and our full identity mod n = (m−1)/2 holds at each of these levels (433a1, c_∞ = 1, is the tenth curve in the table, added for this comparison, with u_E = 4 at 21 levels). The two pairings are related by the projection $G_m^+ \to \mathrm{Gal}(F/Q) \cong Z/3$, so this is the expected agreement rather than an independent confirmation of the values; matching Ψ entry by entry would require re-running their code.

Independent re-implementation and blind test. After this note reached its present form, the prime-level identity was re-implemented from the text alone (without access to our scripts) in a separate session, and the constant $u_E = (-2)^r c_\infty / \#E(Q)\mathrm{tors}^2$ was recorded in advance for seven curves not in the table above. The identity then held on all of them — 563a1, 643a1, 664a1 (c_Tam = 4), 681c1 (c_Tam = 2), 707a1 (c_Tam = 2), 997b1, and the rank-3 curve 11197a1 — at 237 non-trivial levels (7 ≤ m ≤ 200 in rank 2, m ≤ 150 in rank 3) with no failures, u_E as predicted in every case, including the sign −16 for the second rank-3 curve. Negative controls fail as they should: a wrong u_E, omitting c_Tam, pairing the points of one curve with the theta element of another, or the wrong sign in rank 3 each give at most 2 accidental agreements out of 26–39. The re-implementation also reproduces the Appendix exactly. Both implementations use PARI/GP 2.17.2, so this is independence of code and of the choice of constant, not of the library.

The same re-implementation, with the hypothesis $u_E = -2c_\infty$ recorded in advance, verified the rank-1 case on 37a1, 43a1, 53a1, 57a1, 58a1, 61a1, 77a1 and 79a1 at 533 levels m ≤ 400 with no failures (the opposite sign agrees at 1–3 levels per curve, by chance); all eight curves have trivial torsion, so the general law reduces there to $u_E = -2c_\infty$. In rank 0 the identity is the statement about the augmentation class $c_0(m) = \Sigma\, c_t$, and the law takes the form $u_E \cdot \#E(Q)\mathrm{tors}^2 = c_\infty \cdot$ #Ш (hypothesis recorded in advance); it held at all 249 levels m ≤ 400 on eight curves, including the two with non-trivial Ш (571a1, #Ш = 4; 681b1, #Ш = 9):

Curve	c∞	c_Tam	#E(Q)tors	#Ш	levels	failures
11a1	1	5	5	1	29	0
14a1	1	6	6	1	18	0

Curve	c∞	c_Tam	#E(Q)tors	#Ш	levels	failures
15a1	2	8	8	1	30	0
32a1	1	4	4	1	39	0
67a1	1	1	1	1	39	0
75a1	1	1	1	1	27	0
571a1	1	1	1	4	37	0
681b1	2	4	4	9	30	0

The 67 levels of 571a1 and 681b1 are the only ones, in this note or in the re-implementation, that test the factor #Ш of the conjecture with #Ш > 1; the other 182 rank-0 levels test the case #Ш = 1, already covered in positive rank. With #Ш included, the constant is $u_E = (-2)^r \cdot c\infty \cdot \#Ш / \#E(Q)tors^2$ in ranks 0–3. In all, the identity has been checked on 33 distinct curves of ranks 0–3 at 1722 non-trivial prime levels (703 in this note, 237 + 533 + 249 in the re-implementation, not counting its re-check of the ten curves of the table).

Rank 2 with #Ш > 1 (exploratory). Since no optimal rank-2 curve of conductor ≤ 500,000 has #Ш > 1, the factor #Ш in positive rank was tested separately on four rank-2 curves found in the LMFDB: 53295337.a1 and 136116413.a1 with #Ш = 9 (trivial torsion, c_Tam = 1), and the non-optimal curves 499290e4 and 226005g3 with #Ш = 4. At these conductors PARI's msfromell is out of reach, so θ_m was computed from the twisted L-values L(E, χ, 1) instead (validated coefficient by coefficient against msfromell on 425 pairs (E, m) with N ≤ 5077, covering c∞ = 1 and 2). With the constant inferred from the data alone, the identity held with no failures and the inferred constant was exactly $4 \cdot c\infty \cdot \#Ш$ (36 and 72) on the two #Ш = 9 curves, while the law without #Ш fails at every discriminating level; one of them, 53295337.a1, was reproduced independently (PARI's lfun/lfuntwist, separate code) at its 4 discriminating levels m = 11, 23, 29, 31, 4/4 with #Ш and 0/4 without. The two #Ш = 4 curves give a weaker test: $\#Ш/\#E(Q)tors^2$ is 1 and 1/4 there, so the data do not separate #Ш from the torsion factor, and the even torsion loses 2-primary information. To our knowledge this is the first test of the #Ш factor of the refined conjecture in rank ≥ 2 — a claim to be confirmed by a specialist; the computation is small (four discriminating levels on one curve reproduced) and is reported as exploratory.

5. Direction of Kolyvagin classes in rank 2

Let E/Q have rank 2 and root number +1, K an imaginary quadratic field satisfying the Heegner hypothesis for N, and p an odd prime with E[p] irreducible, $p \nmid N \cdot c_{Tam} \cdot \#E(K)tors$. For a Kolyvagin prime ℓ (inert in K, $p \mid a_\ell$, $p \mid \ell+1$) let $y_\ell \in E(K[\ell])$ be the Heegner point of conductor ℓ and $\kappa_\ell \in H^1(K, E[p])$ the Kolyvagin derived class of $\Sigma\, \sigma\, D_\ell\, y_\ell$, the sum over coset representatives σ of $G = Gal(K[\ell]/K[1]) \cong Z/(\ell+1)$ in $Gal(K[\ell]/K)$. Since the Heegner point $y_K \in E(K)$ is torsion (here literally 0), $\kappa_\ell \in Sel_p(E/K)^+ = E(Q)/pE(Q) \cong F_p^2$ when $Ш(E/Q)[p] = 0$. Let Q generate $E^D(Q) \otimes Q$, D = disc K; in every case below rank $E^D(Q) = 1$, checked with ellrank.

Statement tested. (a) $\kappa_\ell \neq 0$ if and only if the reduction map $E(Q)/p \to E(F_\ell)/pE(F_\ell)$ is nonzero and Q is not p-divisible in $E^D(F_\ell)$. (b) When $\kappa_\ell \neq 0$, $F_p \cdot \kappa_\ell = \ker(E(Q)/p \to E(F_\ell)/pE(F_\ell))$; in particular the line is independent of K.

Both follow from the transverse local condition at ℓ in Kolyvagin systems (Howard 2004, Def. 1.2.2; Mazur–Rubin 2004) together with the vanishing of y_K and the core-rank-one structure theory; the statement appears as Prop. 0.2 of an unpublished and incomplete draft hosted on W. Stein's website (no author or date given; its proof refers to a missing lemma). We are not aware of any numerical test of (b).

Method. The $h = \ell+1$ (or $2(\ell+1)$ when h(K) = 2) Heegner points of conductor ℓ are computed to 150–450 digits from the modular parametrization; $x(y_\ell)$ has a rational minimal polynomial f of degree h, recognized exactly (denominators up to 10^{113}). A reduced primitive element $\theta = B(x)$ (`polredbest`) gives $L = Q(\theta)$; the Galois action of a generator σ of G on the roots is recovered, without interpolation, from the integer resolvents $\prod(T - (\theta_k + c(\theta_{\sigma k} + \theta_{\sigma^{-1}k})))$, c = 1,2,3, and (for orientation) the O_K-resolvents $\prod(T - (\theta_k + c\theta_{\sigma k}))$, c = 2,3. The y-coordinate is recovered from the exact square root of $(4x^3 + b_2x^2 + 2b_4x + b_6)/(4D)$ in L (`nfroots`), whose existence encodes the action of complex conjugation. At each prime $q \equiv 1 \pmod p$ totally split in $K[\ell]$, all points reduce to $E(F_q)$, the derivative $D_\ell = \Sigma\, i\sigma^i$ is applied along the σ-cycle of roots mod q, and the class of the result in $E(F_q)/p$ is read off by the order-p Tate pairing (checked against direct enumeration of $pE(F_q)$) and compared with the reductions of P_1, P_2 and Q. Twenty such primes identify the class uniquely; the trace relation $\mathrm{Tr}\, y_\ell = a_\ell\, y_1$ holds at every prime as a check. A control case with a theorem-level prediction (43a1, rank 1, $K = Q(\sqrt{-8})$, $\ell = 29$, where κ = 0 is forced) gave 0 at 15/15 primes.

Curve	p	K	ℓ	predicted	computed
389a1	3	$Q(\sqrt{-5})$ (h = 2)	17	$\langle P_1 - P_2 \rangle$	$-(P_1 - P_2)$
389a1	3	$Q(\sqrt{-7})$	5 / 17 / 41 / 59	$\langle P_1+P_2 \rangle$ / 0 / $\langle P_1-P_2 \rangle$ / $\langle P_2 \rangle$	all as predicted
389a1	3	$Q(\sqrt{-11})$	17 / 29 / 41	$\langle P_1-P_2 \rangle$ / 0 / $\langle P_1-P_2 \rangle$	all as predicted
389a1	3	$Q(\sqrt{-19})$	29 / 41 / 53 / 59	$\langle P_1-P_2 \rangle$ / $\langle P_1-P_2 \rangle$ / 0 / $\langle P_2 \rangle$	all as predicted
389a1	**5**	$Q(\sqrt{-7})$, $Q(\sqrt{-11})$	19	$\langle P_1+P_2 \rangle$	as predicted (both)
571b1	3	$Q(\sqrt{-7})$ / $Q(\sqrt{-8})$	47, 59 / 47	0, $\langle P_1+P_2 \rangle$ / $\langle P_1+P_2 \rangle$	all as predicted
709a1	3	$Q(\sqrt{-7})$	5 / 47 / 59	$\langle P_1 \rangle$ / 0 / $\langle P_1-P_2 \rangle$	all as predicted

Twenty cases, three curves, two primes p, five zero predictions, four distinct lines of F_p^2, no failures (the 43a1 control is not counted). Basis vectors P_1, P_2 are the saturated `ellrank` generators of each curve (for 389a1 in §5: (−1,1), (0,0) in the first row, `ellrank`'s (0,−1), (1,0) elsewhere; since (0,−1) = $-P_2$ and (1,0) = $P_1 + P_2$ in the first basis, $P_1 - P_2 \equiv P_2' - P_1'$ (mod 3) and the label $\langle P_1 - P_2 \rangle$ denotes the same line in both). For h(K) = 2 the class must be summed over the two cosets of G with consistent orientation; summing one coset gives a well-defined but wrong class, which is how the h = 2 computation first failed.

Comparison with Jetchev–Lauter–Stein. Two of the cases above were computed before, by a different method: JLS (2009, Prop. 3.10 and Rem. 3.11) show $\kappa_5 \neq 0$ for 389a1 and for 709a1 with $K = Q(\sqrt{-7})$, $p = 3$, by proving that the Heegner point P_5 is not 3-divisible in $E(K[5])$ (the polynomial $\varphi_3(x) - x(P_5)\psi_3(x)^2$ has no root in $K[5]$). Our table gives $\kappa_5 \neq 0$ in both cases, with the additional information of the line: $\langle P_1+P_2\rangle$ for 389a1 and $\langle P_1\rangle$ for 709a1. JLS did not compute the direction of the class, and the two methods share no code or intermediate data beyond the Heegner points themselves; their nonvanishing results and ours agree.

Two-legged classes in rank 3 (exploratory). For 11642a1 (rank 3), $K = Q(\sqrt{-7})$, $p = 3$ and $n = 5\cdot17$, the derived class $\kappa_n \in E(Q)/3 \cong F_3^3$ was computed from Heegner points of degree 108 over K, in an isolated session. Before the computation the prediction that κ_n lies in $\ker \mathrm{red}_5 \cap \ker \mathrm{red}_{17} = \langle P_1 + 2P_2 + 2P_3\rangle$ (or vanishes) was recorded. The class is non-zero and spans exactly this line at all 25 auxiliary primes q, and $\kappa_5 = \kappa_{17} = 0$ as forced; a recomputation at 2600 digits with a different recognition method gave identical output. That κ_n lies in the intersection follows almost formally from the Kolyvagin-system axioms; the non-trivial content is $\kappa_n \neq 0$ for this pair. It is a single case, and κ_n was not recomputed by independent code. We know of no earlier explicit computation of a two-legged class in rank 3, but have not searched the literature systematically.

6. Relation to the literature and what is new

Theory. Mazur–Tate (1987) formulated the refined conjectures; we cite it only through Tan, Ota, Burns–Kurihara–Sano and Portillo-Bobadilla, not having had access to it. Ota (2018) proved $\theta_S \in I^{r}$ (the order part) for the p-parts under hypotheses, and the implication "leading term $\neq 0$ mod p $\Rightarrow$ Ш[p] = 0 and p ∤ J", both for $p \geq 5$ (his §§3, 5). Burns–Kurihara–Sano (2021, published IMRN 2025) recast the leading-term conjecture in the form used here and proved cases in rank 1 (Cor. 1.2: supersingular reduction, squarefree conductor and further conditions, the leading term up to an automorphism of Gal(F/Q)); they also remark (§1) that the only supporting numerical evidence then available was for r = 1 (Portillo-Bobadilla); Bullach–Honnor (2025) prove substantial parts of Mazur–Tate's Conjectures 3, 4 and 6 for $p > 3$ under a big-image hypothesis (e.g. E semistable and $p \geq 11$) and mild local conditions; their leading-term result is trivial in positive rank (their Rem. 1.8(b)). Bertolini–Darmon (1994, 1995) introduced derived heights for the degenerate case; the parity phenomenon in §4 is consistent with their Thm. 3.18.

Numerics. Portillo-Bobadilla (2019) treats rank 1 at primes of good reduction, computing the pairing by his eq. (18) (the denominator cross-ratio with an auxiliary point of Mazur–Tate 1987, p. 747, which agrees with our formula, §2), on the first 3000 curves of Cremona's tables and on 20 curves with #Ш = 4 and 7 with #Ш = 9 — so the factor #Ш > 1 had been tested in rank 1 before; the rank-0 and rank-2 tests of §4 extend this, not the rank-1 case. Séguin (McGill M.Sc. thesis, 2014/15) tests Darmon's Heegner-point analogue of the refined conjecture, in conductor 37 over real quadratic fields, computing Mazur–Tate pairings in that setting; it is not a test of the Mazur–Tate conjectures over Q. Llerena-Córdova (2024) tests three of Mazur–Tate's conjectures on 425,713 pairs (E, p) with split multiplicative reduction at p, finding failures that disappear when a factor $1/\#E(Q)_{tors}$ is included; it does not treat the rank ≥ 2 leading-term statement with a Mazur–Tate regulator (its Conj. 2.12 asserts only the order of vanishing in positive rank); its conclusion that $1/\#E(Q)_{tors}$ must be adjoined is consistent with the factor $1/\#E(Q)_{tors}^2$ in u_E found here. The closest work is Bley–Macías Castillo (2022): a numerical verification, assuming classical BSD for E/F and Ш[3] = 0, of the 3-adic equivariant Tamagawa number conjecture — equivalent to congruences in the augmentation filtration involving the Mazur–Tate pairing — for 389a1 (ℓ = 7, 13, 43), 433a1 and 446d1 (ℓ = 19), in the cubic subfield of $Q(\zeta_\ell)$, with the pairing computed via generalized Selmer groups and Hilbert symbols in Magma.

Statistics of det H mod p (exploratory). On a further sample of 15,034 curves of ranks 1–3 (about 3.7 million curve–level pairs; scripts, registered hypotheses and summary tables in the companion repository, raw data on request), the reduction of the pairing matrix H mod p behaves like a random symmetric matrix over F_p (Haar model), with one exception registered as a hypothesis before the relevant data were collected: for curves with a rational 3-isogeny and no rational 3-torsion, $\det H \equiv 0 \pmod 3$ is more frequent than the model predicts — in rank 2, 0.374 against 1/3 on 334 curves ($z \approx 6.7$, $p \approx 10^{-8}$); in rank 3 only 5 curves, 0.49 against 29/81; no effect in rank 1, as predicted. Part of the excess is explained by the isotropy of the image of the dual isogeny; a corrected mechanism with local terms at $\ell \mid 6N$, found after seeing the data, accounts for the rest in the cases examined and is left as a prediction to be tested on new data. The statistics were recomputed from the raw files by the registration session.

What is new here, to the best of our knowledge: (i) a verification of the leading-term conjecture in rank 3; (ii) in rank 2, a verification with θ_m computed from modular symbols at the full level G_m^+ (moduli (m−1)/2 up to 749), with both sides independent, at 21–221 non-trivial levels per curve; (iii) the auxiliary-point-free closed formula for the pairing of §2 — $\langle R,S\rangle = [\gamma/A(R+S)]$, $\langle R,R\rangle = [A(R)]^{-1}$ — with its derivation from the biextension and the integrality Lemma; the pairing was already computable from coordinates by the denominator cross-ratio with an auxiliary point of Mazur–Tate (1987, p. 747), used by Portillo-Bobadilla (2019, eq. (18)), with which ours agrees numerically, so the novelty is the closed form and the Lemma, not the possibility of computing the pairing from coordinates; (iv) the explicit confirmation that the Tamagawa factor enters through the joint kernel defining $E_S(Q)$; (v) the direction test for Kolyvagin classes of §5. Items (i)–(iv) are verifications of an existing conjecture, not new mathematics; item (iii) is a formula whose proof is complete except for the identification of the local splittings named in §7.1.

7. Caveats, open points and next steps

1. **Status of the pairing formula.** §2 derives the recipe from the biextension definition, granting the Néron-model description of the canonical local splittings (vertical order at unramified places; m-adic order and reduction to the special fibre at m). The elementary integrality statement used there — that the cancellation δ in the duplication formula is 1 for points of $E_0(Q_l)$ — is now proved for every prime ℓ, $\ell = 2$ included (Lemma in §2): in fact $\ell \mid \delta$ if and only if $R \notin E_0(Q_l)$, by an integral identity that avoids the formal group; it is also consistent with the 349 + 3631 points checked earlier, which cover every reduction type at 2. What remains is for someone with Mazur–Tate (1983) in hand to confirm that the splittings we used are theirs; Tan's description of (L3)/(L4) and the uniqueness argument of §2 (iii) leave little room for a different choice. With that confirmed, the formula $\langle R,S\rangle = [\gamma_{\{R,S\}}/A(R+S)]$, and hence $\langle R,R\rangle = [A(R)]^{-1}$, is a theorem.
2. **Powers of 2 and sign — explained.** The constant $u_E = (-2)^r \cdot c_\infty / \#E(Q)_{tors}^2$ of §4 is explained factor by factor, sign included, and the stress tests on 655a1, 1132a1, 916c1 and 1088j1 confirm that the Tamagawa numbers do not enter it. Our conventions are: $x = \sigma - 1$ (so $\eta_r(g_1 \cdots g_r) =$

$\prod(\sigma(g_i) - 1)$, as in Ota and Burns–Kurihara–Sano), the primitive root g fixed by PARI's znprimroot, PARI's plus symbol (which has $x^+(\{0\}-\{\infty\}) = -L(E,1)/\omega_1$, the opposite sign to Mazur–Tate's $[0]^+ = L(E,1)/\Omega^+$), and the pairing $\langle R,S\rangle = [\gamma/A(R+S)]$ of §2, whose inverse is the pairing in Tan's orientation. Undoing these two sign conventions turns our $(-1)^r$ into Ota's $(-1)^{v(m)} = -1$ for every r, as §4(b) explains, assuming that Ota normalizes the Artin map as Tan does (uniformizer ↦ arithmetic Frobenius; the standard choice, not checked in his text and asked of him directly); the rank-1 data of the independent re-implementation (533 levels, $u_E = -2c\infty$) were what fixed the dependence on r. The v0.1 text of this item said the sign was open and suggested a factor $(-1)^{r(r-1)/2}$; both statements were wrong.

3. **Weak levels.** When gcd(M, n) > 1 the identity determines $c_r(m)$ only modulo n/gcd(M,n). About half the levels for 389a1 are of this kind. They are still non-trivial checks, but the headline counts should be read with this in mind.
4. **One analytic input.** #Ш = 1 is taken from the analytic order (strong BSD), and every curve of rank ≥ 1 tested here (ours and the re-implementation's) has #Ш = 1, so the factor #Ш in the conjecture is tested only in rank 0 (by the re-implementation, on 571a1 with Ш = 4 and 681b1 with Ш = 9; Llerena-Córdova's rank-0 tests cover it on a much larger scale at primes of split multiplicative reduction). The identity tested in positive rank is therefore a joint consistency check of Mazur–Tate and of BSD's Ш-term; a curve with Ш ≠ 1 in rank 2 would be a sharper test (among the 279,056 optimal rank-2 curves in Cremona's tables up to conductor 500,000 the analytic #Ш is always 1, by the allbsd column, checked with an awk one-liner recorded in the companion document; non-optimal curves in the same isogeny classes can have #Ш = 4, e.g. 499290e4 and 226005g3, and rank-2 curves with #Ш = 9 exist at larger conductors, e.g. 53295337.a1 and 136116413.a1 in the LMFDB, so such a test is possible and is left for future work).
5. **Composite levels** are now tested through Ota's inclusion–exclusion formula for $S = m_1m_2$ (§4); the mixed coefficient x_1x_2 is the only genuinely new information at composite level and is checked only modulo $\gcd(n_1, n_2)$, which limits the strength of each individual check; extending to three prime factors and to larger gcd is straightforward but costly (the points R_i have $\sim 10^6$ digits).
6. **Rank 4** (234446a1, [1,−1,0,−79,289], $c_{Tam} = 2$, $c_\infty = 2$). The code runs unchanged for r = 4 (the identity is in I^4/I^5 with a 4×4 determinant), but PARI's `msfromell` for conductor 234446 needs more than the 7 GB available to us: the attempt was killed by the out-of-memory handler before the first level. On a machine with ∼32 GB, or with a modular-symbol implementation that works in the plus quotient only, this is a few hours of computation.
7. **Bley–Macías Castillo.** The comparison in §4 is at the level of $\det \Psi \neq 0 \pmod 3$, which is all their published output allows. An entry-by-entry match of the 2×2 matrix Ψ against the mod-3 reduction of our pairing matrix would be a sharper test; it requires running their Magma code (available on W. Bley's page), to which we have no access.

None of this touches the archimedean Birch–Swinnerton-Dyer conjecture: the "derivative" in the Mazur–Tate conjecture is in the group-ring direction σ − 1, not in the complex variable s, and the identification of the two is the open problem.

Appendix. A worked example: 389a1 at m = 23

E: $y^2 + y = x^3 + x^2 - 2x$, $P_1 = (-1, 1)$, $P_2 = (0, 0)$, $c_{Tam} = 1$, $c_\infty = 2$, $E(\mathbb{Q})_{tors} = 0$. Take m = 23, so n = 11, g = 5 (PARI's primitive root), $G_{23}^+ = \langle\sigma\rangle \cong \mathbb{Z}/11$.

Theta element. The plus symbols $[a/23]^+$ for a = 1, …, 22 are

0, −1, 0, 0, −1, −1, −1, 1, 2, 1, 0, 0, 1, 2, 1, −1, −1, −1, 0, 0, −1, 0

(symmetric under $a \leftrightarrow 23 - a$, as they must be). Collecting a and −a by $t = \log_5(a) \bmod 11$, the coefficient vector of $\theta_{23} = \Sigma\, c_t\, \sigma^t$ is

$(c_0, \ldots, c_{10}) = (0, -2, -2, 2, 0, 0, 2, -2, -2, 0, 4)$.

Expanding $\sigma^t = (1 + x)^t$: $a_0 = \Sigma\, c_t = 0$, $a_1 = \Sigma\, t\, c_t = 22 = 2\cdot 11$, $a_2 = \Sigma\, C(t,2)\, c_t = 116$. So $\theta_{23} \in I$, and since $a_1 = 2n$ the relation $n\cdot x \equiv -C(11,2)x^2$ gives $\theta_{23} \in I^2$ with leading class $c_2(23) = 116 - 2\cdot 55 = 6 \in \mathbb{Z}/11$.

Points. $\#E(F_{23}) = 28$ ($a_{23} = -4$), so M = 28 and $R_i = 28\cdot P_i$. The x-coordinate of R_1 has a 234-digit numerator and denominator $d(R_1)^2$ with

$d(R_1) = 607852338707617298241916768285023402187465773080649173756581131907387799193727318114990310685196156742789291848564509$ (117 digits),

$d(R_2) = 161961604235457625196184711884753920723064530210946525 77$ (56 digits), and $d(R_1 + R_2)$ has 81 digits. In all three cases $v_{23}(d) = v_{23}(t) = 1$, as expected for points in $E_1(\mathbb{Q}_{23}) \setminus E_2(\mathbb{Q}_{23})$. The unit parts give

$(t/d)(R_1) \equiv 14$, $(t/d)(R_2) \equiv 14$, $(t/d)(R_1 + R_2) \equiv 20 \pmod{23}$,

and with $5^{10} \equiv 9 \equiv -14$, $5^5 \equiv 20 \pmod{23}$ the logarithms in $\mathbb{Z}/11$ are $h(R_1) = 10$, $h(R_2) = 10$, $h(R_1+R_2) = 5$. (That $h(R_1) = h(R_2)$ is a coincidence of this level.) The polarized matrix is

$H = (2h(R_1),\ h(R_1+R_2) - h(R_1) - h(R_2);\ \cdot,\ 2h(R_2)) = (20, -15; -15, 20) \equiv (9, 7; 7, 9) \pmod{11}$, $\det H \equiv 81 - 49 \equiv 10 \pmod{11}$.

The identity. Left side: $2^2\cdot M^3\cdot c_2 = 4\cdot 28^3\cdot 6$; with $28 \equiv 6$ and $6^3 \equiv 7 \pmod{11}$ this is $4\cdot 7\cdot 6 = 168 \equiv 3$. Right side: $u_E\cdot\det H = 8\cdot 10 = 80 \equiv 3 \pmod{11}$. The two sides agree, with the same $u_E = 8 = (-2)^2\cdot c_\infty$ as at the other 220 non-trivial levels for this curve. In the normalization of §4 ($\theta^+ = \theta/c_\infty$ has leading class 6/2 = 3; torsion trivial) this reads $M^{-3}\cdot\det H = 8\cdot 10 \equiv 3 \equiv \theta_{23}/c_\infty$; with the Mazur–Tate sign of the symbol ($\theta^+ = -\theta/c_\infty$, class $-3 \equiv 8$) and Tan's orientation of the pairing ($P = -H$, $\det P = \det H$), this is $\theta_{23}^+ \equiv -J_{23}\cdot \mathrm{disc}_{23}(P)$ in $I^2/I^3 \cong \mathbb{Z}/11$, i.e. Ota's statement with $(-1)^{v} = -1$.

The whole computation is `thetaclass(E,23,2)` and `pairdet(E,23,[P1,P2],28,11,Mod(5,23))` in `mt_rank.gp`, and takes well under a second.

8. Reproducibility, acknowledgements, references

Code and data. All computations use PARI/GP 2.17.2 through cypari2 2.2.4, on Cremona's tables (`allbsd`, `allcurves`). The scripts `mt_full.gp` / `mt_full_run.py` (prime-level identity, rank 2), `mt_rank.gp` / `mt_rank_run.py` (general rank), `mt_lead.gp` (mod-3 version), `mt_comp.gp` (exploratory composite levels), mt_ota.gp / mt_ota_run.py (Ota's composite-level formula), biext.gp / chkg2.gp / chord.gp / gsweep.py (the biextension computation of §2 and the integrality sweep), portillo.gp (comparison with Portillo-Bobadilla's eq. (18)), `heeg.gp` + `kgen3.py` (Kolyvagin classes, any curve and p) and the raw logs are in the accompanying archive. A typical prime-level check for 389a1 up to m = 1500 takes about 40 minutes on one core; the Kolyvagin computation for a field of degree 60 takes 15–20 minutes, dominated by `polredbest`. Two PARI conventions matter: `elltatepairing(E, P, Q, m)` takes the m-torsion point as **P**, and its value lies in $\mathrm{F_q}^x/(\mathrm{F_q}^x)^m$ and must be raised to (q−1)/m; `ellpadicheightmatrix` returns [A, B] with the cyclotomic height matrix equal to $A - s_2 \cdot B$.

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